

Thomas KURMANN, PHD

Staff Machine Learning Engineer | Robotic Perception & Real-Time ML

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Swiss and Canadian citizen. Languages: English and German (fluent), French (conversational).

Staff Machine Learning Engineer (PhD) building production-grade real-time computer vision for robotic surgery. Specialized in stereo depth estimation, multi-task perception, multi-spectral imaging, and efficient edge AI under strict latency and robustness constraints. Led the end-to-end delivery of production 3D perception for the da Vinci platform, from synthetic data generation and model development through optimized on-device deployment, supporting FDA 510(k) clearance activities. Expertise in stereo vision, multi-view geometry, and embedded ML with applications spanning robotics, AR/3D, and autonomy. Author of 6 peer-reviewed publications (5 first-author), including MICCAI, Nature Scientific Reports, and IJCARS. Recipient of the MICCAI Young Scientist Award and EndoVis Challenge 1st (2015) and 2nd (2017).

WORK EXPERIENCE

Today 2021	Staff ML Engineer, INTUITIVE SURGICAL, Aubonne, Switzerland Staff ML Engineer (Apr 2025–Present) Senior ML Engineer (Apr 2023–Apr 2025) ML Engineer (Mar 2021–Apr 2023) Intern (Apr–Aug 2018, Sunnyvale, CA) - DRI for stereo matching company-wide: owned the full pipeline end-to-end (synthetic data generation -> training -> validation -> optimization -> deployment -> V&V) and shipped production stereo depth for a da Vinci product, supporting FDA 510(k) clearance activities - Deployed two stereo models to production (high-accuracy and real-time variants) on edge GPU and Jetson hardware, using TensorRT quantization for live in-vivo inference on da Vinci - Designed a shared-backbone multi-task model (stereo, segmentation, classification, optical flow) that cut inference time 30% and reduced model footprint versus separate networks - Built the graph-based inference platform adopted by all ML teams at Intuitive, and an evaluation platform for regulatory-compliant V&V - Engineered the data engine for training and test sets: custom structured-light hardware for ground-truth depth, porcine and in-human experiments, endoscope/calibration limit characterization, and a reverse image search tool for surgical-video curation - Shipped real-time surgical-instrument segmentation and intraoperative anomaly detection on live endoscopic video - Built an autonomous CUDA optimization system using LLM agents that automatically optimized, compiled, profiled, benchmarked and validated kernels, improving production kernels by >=20% - Led a team of 2 engineers and mentored 2 interns; drove close cross-site collaboration with R&D teams in Sunnyvale, CA and Germany Computer Vision Segmentation Multi-Task Learning Stereo Matching Multi-Spectral Imaging Quantization LLM Agents CUDA Edge Deployment
2021	PhD Candidate, AI IN MEDICAL IMAGING LABORATORY, ARTORG CENTER, UNIVERSITY OF BERN, Bern, Switzerland
2016	- Developed deep learning methods for surgical instrument recognition, pose estimation, and segmentation in minimally invasive surgery video - Built expert-level biomarker detection and segmentation models for retinal OCT, achieving clinical-grade performance in ophthalmology - Published 6 peer-reviewed papers (5 first-author, 1 co-first) in top-tier venues (MICCAI, Nature Scientific Reports, IJCARS); MICCAI Young Scientist Award (2019) and Prix Retina (2020) - Supervised by Prof. Dr. Raphael Sznitman, with collaborations from Prof. Dr. Sebastian Wolf (University of Bern) and Prof. Dr. Greg Hager (Johns Hopkins University) Computer Vision Deep Learning Medical Imaging Ophthalmology
2015 2008	Earlier Career, SOFTWARE, ELECTRICAL & MECHANICAL ENGINEERING, Switzerland - Software, embedded, and electrical engineering across regulated/industrial settings: enterprise Java/Spring Boot backends and REST APIs (Comlab AG); embedded Linux and real-time data acquisition for seismic/structural monitoring (GSI GeoSolutions, GeoSIG); metrology-grade accelerometer calibration (Swiss Federal Office of Metrology) Software Engineering Embedded Systems Hardware Design Electrical Engineering Mechanical Engineering C/C++

SKILLS

Robotic Perception	Stereo Matching, Depth Estimation, 3D Reconstruction, Multi-View Geometry, Multi-Spectral Imaging, Camera Calibration, Optical Flow, Pose Estimation
Computer Vision & Deep Learning	Multi-Task Learning, Semantic Segmentation, Object Detection, Vision Transformers, CNNs, Quantization, Model Optimization
Edge AI & Performance	TensorRT, CUDA Kernel Optimization, Embedded Inference, NVIDIA Jetson, Real-Time Systems
Programming	Python, C++, CUDA, C
Frameworks	PyTorch, ONNX, OpenCV
Systems & Tooling	Production ML Pipelines, Synthetic Data Generation, Docker, GitHub Actions, Linux, CMake

PUBLICATIONS

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1. **Kurmann, T.**, Márquez-Neila, P., Du, X., et al. *Simultaneous recognition and pose estimation of instruments in minimally invasive surgery*. MICCAI, 2017. doi:10.1007/978-3-319-66185-8_57, arXiv:1710.06668
 2. **Kurmann, T.**, Márquez-Neila, P., Wolf, S., Sznitman, R. *Deep multi-label classification in affine subspaces*. MICCAI, 2019. doi:10.1007/978-3-030-32239-7_19, arXiv:1907.04563
 3. **Kurmann, T.**, Márquez-Neila, P., Yu, S., et al. *Fused detection of retinal biomarkers in OCT volumes*. MICCAI, 2019. doi:10.1007/978-3-030-32239-7_29, arXiv:1907.06955
 4. **Kurmann, T.**, Yu, S., Márquez-Neila, P., et al. *Expert-level automated biomarker identification in optical coherence tomography scans*. *Nat. Sci. Rep.*, 9:14497, 2019. doi:10.1038/s41598-019-49740-7
 5. Munk, M.R., **Kurmann, T.** (co-first), Márquez-Neila, P., Zinkernagel, M.S., Wolf, S., Sznitman, R. *Assessment of patient specific information in the wild on fundus photography and optical coherence tomography*. *Nat. Sci. Rep.*, 11:8621, 2021. doi:10.1038/s41598-021-86577-5
 6. **Kurmann, T.**, Márquez-Neila, P., Allan, M., Wolf, S., Sznitman, R. *Mask then classify: multi-instance segmentation for surgical instruments*. IJCARS, 16(9), 2021. doi:10.1007/s11548-021-02404-2

EDUCATION

2016 – 2021	PhD in Biomedical Engineering, University of Bern, Switzerland
2013 – 2015	Msc. Biomedical Engineering, University of Bern, Switzerland
2008 – 2011	BSc. Electrical Engineering, University of Applied Sciences Bern, Switzerland
2004 – 2008	EFZ Elektroniker Glattbrugg, Switzerland

HONORS & AWARDS

- Prix Retina , Swiss VitreoRetinal Group	Jan 2020
- MICCAI Young Scientist Award , MICCAI Society	Nov 2019
- EndoVis (MICCAI): 1st place (2015), 2nd place (2017)	
- Eurobot National Champion , Swiss Eurobot Competition	2011